

1 Introduction

For the fourth report, the NacionalnaKlasa team refined the system by completing UDP communication and adding a debugging frontend with multiple windows. We recalibrated steering to fix turning imbalance and extended the state machine with roundabout and overtaking behaviors. We also began exploring integration of newly acquired sensors and additional hardware.

2 Planned activities

During the holiday period, the team remained flexible, contributing whenever possible to ensure continuous project progress:

✚ Konstantin Malešević & Filip Goldberger

Konstantin and Filip focused on extending the state machine by adding the new behaviors for roundabouts and overtaking, as well as completing the UDP communication. They are also exploring the integration of newly ordered hardware components.

✚ Minja Drakul & Iva Mančev

Minja and Iva worked on developing the debugging frontend and are currently researching hardware integration while continuing to improve the existing traffic sign detection model.

3 Status of planned activities

- Vehicle calibration updated with steering center adjustment to fix turning imbalance DONE
- Development and extension of the state machine (including roundabout and overtaking states) DONE
- Implementation of UDP communication for real-time data exchange DONE
- Development of a debugging frontend with multi-window support and streaming over UDP DONE
- Improvement of the custom traffic sign detection model in progress
- Exploration of integration of newly acquired sensors and additional hardware in progress

4 General status of the project

- ✚ Steering calibration was refined by adjusting the center angle, resolving the previously observed imbalance in turning and improving overall control stability.
- ✚ The state-based control system was extended with additional behaviors, including handling of roundabouts and overtaking scenarios, allowing the vehicle to operate in more complex traffic situations.
- ✚ The UDP-based communication architecture enables real-time transmission of video, state data, and telemetry. Messages are split into chunks if needed, ensuring large payloads like video frames are reliably delivered. A FPSClock regulates frame processing and sending, preventing network congestion while maintaining consistent performance. This low-latency, modular setup is critical for debugging, testing, and verifying the autonomous vehicle's behavior under dynamic conditions.
- ✚ A custom debugging frontend was developed using Flask to enable real-time monitoring of the system. It supports multiple video streams and displays the current

vehicle state, allowing us to detect delays, synchronization issues, and overall system performance during runtime by transmitting selected data over UDP.

- ✚ In parallel, work continued on improving the custom traffic sign detection model through further training and optimization.
- ✚ Additionally, new sensors and hardware components have been acquired, and their integration into the existing system is under investigation. The evaluated hardware includes VL53L1X Time-of-Flight (ToF) sensor and ultrasonic sensors for distance measurement comparison. For visual tasks, the system uses the Raspberry Pi Camera Module 3 (Sony IMX708), along with an RGB LED strip and 5mm LEDs for signaling and lighting. Edge AI capabilities are explored using the Raspberry Pi 5 AI HAT+, enabling on-device inference. The AS5600 magnetic rotary encoder is also being evaluated for precise angular position sensing and motion tracking.

5 Upcoming activities

In the next period, we plan to work on:

- ✚ Improve the traffic sign detection model through further training, optimization, and testing to enhance accuracy and robustness in real-world conditions.
- ✚ Enhance the state machine by fine-tuning existing behaviors such as roundabout navigation and overtaking, and by adding new dynamic driving scenarios.
- ✚ Integrate new hardware by researching and implementing newly acquired sensors and other components into the vehicle system to expand its capabilities.